

# Phonon-Modulated Hubble Parameter $H(t, \Gamma)$ via $F_U$ and $E_{\text{net}}$ Coupling

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## PAPER\_1085: Phonon-Modulated Hubble Parameter $H(t, \Gamma)$ via $F_U$ and $E_{\text{net}}$ Coupling

### Abstract

We derive the phonon-modulated Hubble parameter:

$$H(t, \Gamma) = H_0 \left(1 + \frac{\Phi}{F_U} \cdot E_{\text{net}}\right)$$

This couples the phonon resonance flux  $\Phi_{1.25 \text{ THz}}$  to the unified gravitational field  $F_U$  and the net vacuum energy  $E_{\text{net}}$ , producing a dynamical Hubble rate that amplifies during periods of strong phonon-vacuum coupling. Unlike PAPER\_1084 (which derives the inflationary  $H$  from  $\rho_{\text{SCm}}$ ), this formulation connects  $H$  to the present-epoch observables  $F_U$ ,  $E_{\text{net}}$ , and  $\Phi$ .

### §1 Core Equation

$$H(t, \Gamma) = H_0 \left(1 + \frac{\Phi(\Gamma)}{F_U(M, r)} \cdot E_{\text{net}}(t)\right)$$

where:

- $H_0 = 2.195 \times 10^{-18} \text{ s}^{-1}$  (67.74 km/s/Mpc)
- $\Phi(\Gamma) = \Phi_0 \cdot S_{26}$  at resonance
- $F_U = \sum_{i=1}^{26} G M / r^2 \cdot [\text{SSq}] \cdot i / 26$  (26-layer gravity)

### §2 Net Vacuum Energy

$$E_{\text{net}}(t) = E_0 \cdot e^{(\kappa + [\text{SSq}]/26) t} \cdot S_{26} \cdot (2R - 1)$$

where  $R = F_{U, Bi} / F_U$  is the buoyancy ratio:

- $R > 0.5$ : expansion (positive  $E_{\text{net}}$ )
- $R < 0.5$ : erosion (negative  $E_{\text{net}}$ )

### §3 Amplification Factor

The phonon amplification of  $H$  is:

$$\frac{H(t, \Gamma)}{H_0} = 1 + \frac{\Phi \cdot E_{\text{net}}}{F_U}$$

For solar parameters ( $M = M_{\odot}$ ,  $r = 1 \text{ AU}$ ,  $R = 0.8$ ):

$H/H_0 \sim 10^6$ , showing enormous amplification — physically this represents the local departure from Hubble flow near massive objects.

## §4 Gamma Dependence

The phonon linewidth  $\Gamma$  enters through  $\Phi(\Gamma)$ :

$$\Phi(\Gamma) = \Phi_0 \cdot \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}$$

At resonance:  $\Phi = \Phi_0 \cdot S_{26} \approx 1.45 \times 10^{21}$ .

Off-resonance:  $\Phi$  decreases Gaussian-sharply with linewidth.

## §5 Physical Interpretation

The formula  $H(t, \Gamma) = H_0(1 + \Phi/F_U \cdot E_{\text{net}})$  encodes:

1. **Cosmic expansion base rate:**  $H_0$  (Hubble constant)
2. **Local phonon enhancement:**  $\Phi/F_U$  ratio (SCm vacuum coupling strength)
3. **Net energy direction:**  $E_{\text{net}}$  sign (expansion vs erosion)
4. **Linewidth modulation:**  $\Gamma$  controls phonon resonance sharpness

## References

- PAPER\_1084: SCm Phonon-Coupled Inflationary Scale Factor
- PAPER\_1073: SCm Phonon-Driven Inflation
- PAPER\_897: PhononModulatedEnergyEnetPhononCalc

### Key References with arXiv/DOI Identifiers

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